

Mathematical Logic Proof Techniques

Exercise Series

Based on Chapter 1: Introduction to Mathematical Logic

Part I: Intermediate (Logic, Tables, and Connectors)

Exercise 1 (Truth Tables for 3 Variables)

Construct a truth table for the following compound proposition. Determine if the statement is a tautology (always true), a contradiction (always false), or a contingency.

$$(P \vee Q) \implies (R \wedge \neg P)$$

Exercise 2 (Translation to Symbols)

Let P : "The server is down", Q : "The website is slow", and R : "Traffic is high". Translate the following English sentences into symbolic logic:

1. The website is slow only if the traffic is high.
2. The server is not down, but the website is slow.
3. If the traffic is high or the server is down, then the website is slow.
4. The website is slow if and only if the server is down and traffic is not high.

Exercise 3 (Logical Equivalence Check)

Using a truth table, verify whether the following equivalence holds:

$$P \implies (Q \vee R) \iff (P \wedge \neg Q) \implies R$$

Exercise 4 (Counterexamples)

For each of the following universal statements, provide a counterexample to prove it is **False**. The domain is the set of real numbers $\mathbb{R}$.

1. $\forall x \in \mathbb{R}, x^2 > x$.
2. $\forall x \in \mathbb{R}, \sqrt{x^2} = x$.
3. $\forall a, b \in \mathbb{R}$, if $a < b$ then $a^2 < b^2$.

Exercise 5 (Negating Basic Quantifiers)

Write the negation of the following propositions in symbolic form, and then translate the negation into clear English.

1. $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x + y = 0$.
2. $\exists x \in \mathbb{R}, x^2 + 1 = 0$.

Exercise 6 (Contrapositive Construction)

Write the contrapositive of the following implications:

1. If n^2 is not divisible by 2, then n is not divisible by 2.
2. If it rains and I have no umbrella, then I get wet.

Part II: Hard (Properties, Nested Quantifiers, Basic Proofs)

Exercise 7 (Algebra of Logic)

Using the fundamental logical equivalences (De Morgan, Distributive, etc.), simplify the following expression as much as possible. Show your steps.

$$\neg(P \vee (\neg P \wedge Q))$$

Exercise 8 (Translating Nested Quantifiers)

Let $L(x, y)$ be the predicate " x loves y ", where the domain for both variables is the set of all people. Translate the following into symbolic logic:

1. Everybody loves somebody.
2. There is somebody whom everybody loves.
3. Nobody loves everybody.

Exercise 9 (Truth Value of Nested Quantifiers)

Determine the truth value of the following propositions where the domain is $\mathbb{Z}$ (integers). Explain your reasoning.

1. $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x + y = 0$.
2. $\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}, x + y = 0$.

Exercise 10 (Direct Proof - Parity)

Prove the following statement using a Direct Proof: "If n is an even integer and m is an odd integer, then $3n + 5m$ is an odd integer."

Exercise 11 (Proof by Contrapositive)

Prove the following statement using Proof by Contrapositive: "Let $x \in \mathbb{Z}$. If $7x + 9$ is even, then x is odd."

Exercise 12 (Proof by Cases)

Prove that for any integer n , the expression $n^2 + n$ is always even. (Hint: Consider the cases where n is even and n is odd separately).

Exercise 13 (Disproving Implications)

Disprove the following implication: "If n is a prime number, then n is odd." (Find a counterexample that satisfies the hypothesis but fails the conclusion).

Exercise 14 (Rationality)

Prove that the sum of a rational number and an irrational number is irrational. (Hint: Use proof by contradiction).

Part III: Advanced (Abstract Proofs, Induction, Formal Logic)

Exercise 15 (The Formal Definition of a Limit)

In calculus, the definition of the limit $\lim_{x \rightarrow a} f(x) = L$ is given by:

$$\forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}, (0 < |x - a| < \delta \implies |f(x) - L| < \epsilon)$$

Write the formal logical negation of this statement (i.e., define what it means for the limit to **not** equal L). Simplify the negation such that no negation symbols appear before a quantifier.

Exercise 16 (Uniqueness Quantifier)

The symbol $\exists!x, P(x)$ means "There exists a **unique** x such that $P(x)$ is true." Rewrite $\exists!x, P(x)$ using only the standard quantifiers $\forall, \exists$, the logical connectors, and the equality symbol $=$.

Exercise 17 (Proof by Contradiction - Logarithms)

Prove by contradiction that $\log_2(3)$ is irrational. (Assume $\log_2(3) = \frac{a}{b}$ and use properties of exponents to reach a contradiction concerning even/odd integers).

Exercise 18 (Proof by Induction - Summation)

Use Mathematical Induction to prove that the sum of the first n squares is given by the formula:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for all integers $n \geq 1$.

Exercise 19 (Proof by Induction - Divisibility)

Prove by induction that for all integers $n \geq 1$, $3^{2n} - 1$ is divisible by 8.

Exercise 20 (Proof by Induction - Inequality)

Prove by induction that $2^n > n^2$ for all integers $n \geq 5$. (Note carefully: The Base Case is $n = 5$).

Exercise 21 (Abstract Contradiction)

Let A and B be sets. Prove the following statement using logical equivalences or contradiction:

$$(A \cap B = A) \iff (A \subseteq B)$$

Recall that $A \subseteq B$ is defined as $\forall x, (x \in A \implies x \in B)$.

Exercise 22 (The Pigeonhole Principle Logic)

Let there be $n + 1$ pigeons and n holes. The statement "at least one hole contains more than one pigeon" is a famous logical truth. Formulate the **negation** of this concept: Describe a scenario where $n + 1$ items are placed into n bins such that no bin contains more than one item, and explain using the contrapositive why this scenario is impossible.